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To the Editors of
Estuarine, Coastal and Shelf Science

Dear Editors,

The manuscript **"Relative effects of seagrass canopy, wave coupling and bed mobility on Lagrangian transport in a shallow lagoon"**, by Cicero Martins Jr., Mauro De Marchis, Giuseppe Ciraolo, Carmelo Nasello, Antonino Maltese, is submitted for consideration as an original research paper. The corresponding author is MSc. Cicero Martins Jr.

Coastal lagoon models are commonly validated against water level and then used to describe transport. The manuscript asks which modelled processes control transport in a shallow vegetated lagoon and whether water level can identify them. A complete factorial in wave coupling, seagrass canopy and bed mobility gives eight configurations of a three-dimensional Delft3D Flexible Mesh model coupled online with SWAN, each scored against three tide gauges and 35 GPS surface drifters over the same nine days.

On the fixed bed the seagrass canopy raises Liu–Weisberg skill by 0.120 and 0.114, with both bootstrap intervals clear of zero, against -0.022 and -0.028 for wave coupling, and with the canopy also dissipating wave energy the latter falls to -0.005. The canopy is represented as a three-dimensional drag field whose parameters come from published measurement of the two species, with nothing fitted. Water level separates the canopy treatments only at the gauge inside the vegetated basin, where anomaly RMSE differs by 26 and 27 mm, against at most 4 mm at the two gauges near the inlets.

The mobile-bed arm shows how an attribution can mislead. With waves, the uncalibrated bed lets a configuration without a canopy reach the skill of the vegetated ones, with near 9 g l^{-1} of suspended sediment in the lagoon against a Sentinel-2 retrieval of a few milligrams per litre. Under silt parameters that hold the bed change at millimetres the bare configuration keeps most of that gain, and the interior gauge exposes the compensation where the drifters do not. A start from a uniform offshore column inflates the canopy effect in the same way, from 0.12 to 0.19.

The work falls within the scope of *Estuarine, Coastal and Shelf Science* as a study of lagoon phenomena and of coastal sedimentary processes. Although the model is built for one basin, the questions it answers concern what a class of validation data can certify and how a factorial behaves when a dominant factor is left out or another is unconstrained, and both apply wherever drifter observations and a circulation model exist together. The prior study closest to the wave treatment, Carniello et al. (2011), appeared in this journal. The limitations are stated in the manuscript, including the uncalibrated morphodynamic arm, kept out of the attribution for that reason, and the single nine-day window.

The corresponding author affirms that:

- All authors have directly participated in the planning, execution, or analysis of this study;
- All authors have read and approved the version submitted;
- The contents of this manuscript have not been copyrighted or published previously;
- The contents of this manuscript are not now under consideration for publication elsewhere.

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The authors remain available for any further clarification the review may require.

Sincerely,

Cicero Martins Jr.
on behalf of all co-authors
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